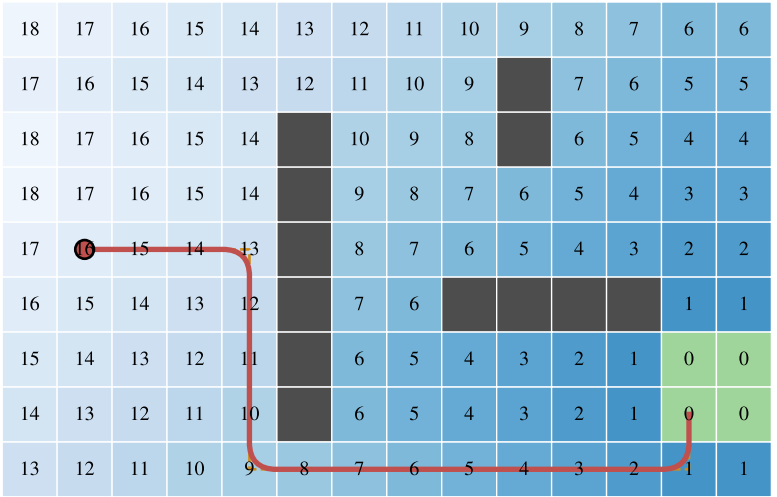
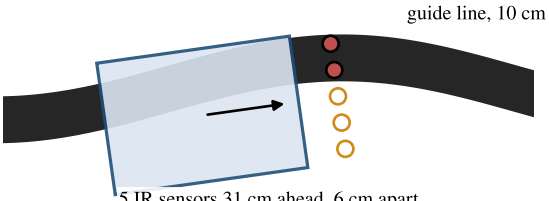
(a) BFS from hidden cells (0); descend; Chaikin ×3 = guide line



(b) PD line follower (top view, metres)



5 IR sensors 31 cm ahead, 6 cm apart

red = over the line: here $e = +0.75$ (turn left)

$e = \overline{\text{offset}}/0.12,\ \omega = \text{clamp}(3.2e + 0.25\,\dot{e},\ \pm 2.5)$

$v = 0.75(1 - 0.6|e|)$ m/s, wheels $v \mp \omega\, w/2$